

Superconductivity in atom-intercalated quaternary hydrides under ambient pressure

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Multinary hydrides offer great promise for achieving high-temperature conventional superconductivity under ambient pressure. However, the rich variety of structural configurations in hydrides poses a formidable challenge for searching their immense phase space. In this work, we used our developed artificial intelligence (AI) search engine (InvDesFlow) to perform extensive investigations regarding ambient stable superconducting hydrides. We assessed the thermodynamic stability of hydrides by our AI model. For K_2GaCuH_6 , the AI-predicted formation energy difference along the synthesis path is 0.146 eV/atom, closely matching the density functional theory result of 0.137 eV/atom. This agreement confirms the thermodynamic stability of hydrides and validates the reliability of AI method. Several quaternary hydrides with high superconducting temperature (T_c) are predicted. In particular, the superconducting T_c of K_2GaCuH_6 and K_2LiCuH_6 are calculated to be 68 and 53 K under ambient pressure, respectively, which shows a significant enhancement in comparison with that of K_2CuH_6 ($T_c \sim 18$ K). We also find that intercalating atoms could cause phonon softening and induce more phonon modes with strong electron-phonon coupling. Hence, we propose that intercalating atoms is a feasible approach in searching for superconducting quaternary hydrides.

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I. INTRODUCTION

Since the first superconductivity was discovered in 1911 [1], research efforts have been focused on finding superconductors with a high critical temperature (T_c) under ambient pressure. In the exploration of conventional superconductors, MgB_2 currently holds the record for the highest T_c among successfully synthesized materials under ambient pressure, with a T_c of 39 K [2], which does not surpass the McMillan limit [3]. According to the Bardeen-Cooper-Schrieffer (BCS) theory [4], lighter elements typically exhibit higher Debye frequencies. Thus, Ashcroft *et al.* proposed that metallic hydrogen could exhibit a high T_c [5]. However, inducing hydrogen into a metallic phase requires extremely high pressures, posing significant experimental challenges [6]. In hydrides, hydrogen atoms are precompressed within crystal lattices, which could lower the extreme environmental conditions needed for the metallization of hydrogen [7]. With the advancement of high-pressure techniques, many high- T_c superconducting hydrides have been reported [8–11].

Early studies primarily focused on binary hydrides, for example, CaH_6 [8] ($T_c = 205$ K at 172 GPa), H_3S [10]

($T_c = 203$ K at 220 GPa), and LaH_{10} [12,13] ($T_c = 250$ K at 170 GPa). However, more systematic exploration has shown that binary hydrides necessitate high pressure to achieve high- T_c superconductivity [14–16], imposing significant constraints on experimental research and practical applications. Therefore, the research focus within the field has expanded to include ternary and multinary hydrides [17–19]. Over the past five years, a series of ternary hydrides have been theoretically predicted to be superconducting. Notable examples include LaBH_8 [20,21], SrSiH_8 , BaSiH_8 [22], a broader class of AXH_8 materials (where $A = \text{Sc, Ca, Y, Sr, La, Ba}$ and $X = \text{Be, B, Al}$) [23], as well as ThBeH_8 and CeBeH_8 [24]. Subsequently, the ternary hydride LaBeH_8 was successfully synthesized under high pressures of 110–130 GPa, exhibiting a superconducting T_c of 110 K at 80 GPa [25]. A similar result has also been reported in La-Ce-H compounds [26].

Recent efforts have identified promising ternary superconducting hydride candidates, such as Mg_2XH_6 ($X = \text{Rh, Ir, Pd, Pt}$) [27] and double perovskite hydrides [28]. Among them, Mg_2IrH_6 was predicted to be superconducting under ambient pressure with a T_c of ~ 160 K [29]. Using the artificial intelligence (AI) search engine, InvDesFlow, our recent work proposed a 216-type ternary hydride Li_2AuH_6 with a superconducting T_c of 140 K under ambient pressure [30]. The strong electron-phonon coupling (EPC) of Li_2AuH_6 is mainly contributed by the Au-H octahedron, which is hence proposed to be a BCS superconducting unit. Further studies regarding Li_2AuH_6 suggested that intercalating extra atoms

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may further enhance EPC, which motivates us to try to search for superconducting quaternary hydrides. Meanwhile, recent relevant reports have validated the efficacy of AI technology in this field, successfully predicting 15 superconducting phases under ambient pressure in quaternary hydride double perovskites [31]. The rising AI technology just provides a suitable approach to investigate quaternary hydride.

In this work, using our AI search engine, InvDesFlow [32,33], we propose atom-intercalated quaternary hydrides based on the 216-type ternary hydrides, where the intercalating elements are selected autonomously by the AI. Subsequently, with the assistance of our AI model, we investigate the thermodynamic stability of these multinary hydrides. The detailed workflow of our AI method with a process diagram is provided in Supplemental Material (SM) [34] Sec. I, and the specific setting of the AI model can be found in the source code [35]. By performing density functional theory (DFT) [36,37] and superconducting EPC studies, we find that the intercalated atoms bring modulation for phonon, EPC, and superconducting T_c of quaternary hydrides. Typically, the superconducting T_c of K_2GaCuH_6 and K_2LiCuH_6 are respectively predicted to be ~ 68 and 53 K under ambient pressure, which is significantly enhanced in comparison with that of K_2CuH_6 (18 K).

II. METHODS

In our calculations, we performed first-principles electronic structure calculations with the QUANTUM ESPRESSO package [38]. The exchange and correlation functional was treated with the Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) [39], while electron-ion interactions were modeled using norm-conserving Vanderbilt pseudopotentials [40]. For all multinary hydride materials, we adopted a plane-wave kinetic energy cutoff of 80 Ry and a charge density cutoff of 320 Ry. The charge densities were calculated on an unshifted $16 \times 16 \times 16$ $\mathbf{k}$ -point grid. The Methfessel-Paxton smearing method [41] was applied with an energy width of 0.02 Ry. We set the following convergence criteria as 1×10^{-4} for total energy and forces in ionic minimization, 1×10^{-16} for self-consistency, and 0.05 kbar for pressure during variable cell relaxation. According to density functional perturbation theory [42], the dynamical matrices and perturbation potentials were evaluated on a Γ -centered $4 \times 4 \times 4$ $\mathbf{q}$ -point grid. A $4 \times 4 \times 4$ $\mathbf{k}$ -point grid in the Brillouin zone was used to construct the maximally localized Wannier functions (MLWFs) [43]. The Electron-Phonon Wannier (EPW) package [44] was employed to evaluate the EPC constant λ through a convergence test within fine electron ($64 \times 64 \times 64$) and phonon ($16 \times 16 \times 16$) grids. A Gaussian smearing of 0.5 meV was applied to the phonon Dirac δ functions. For the electron Dirac δ functions, the specific Gaussian smearing parameters for each hydride were determined through convergence tests, as detailed in SM Sec. II [34]. Ultimately, the T_c of each crystal was obtained by solving the anisotropic Eliashberg equations [44–46]. The Matsubara frequencies were truncated at eight times the maximum phonon energy in each material.

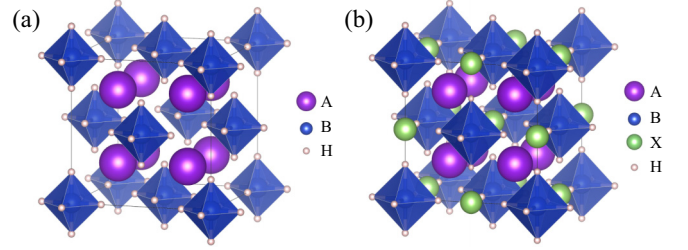


FIG. 1. Crystal structure of (a) A_2BH_6 ($A = K$ and Na ; $B = Cu$ and Ag), (b) A_2XBH_6 ($A = K$ and Na ; $B = Cu$ and Ag ; $X = Ga$ and Li), respectively.

The mode- and wave-vector-dependent coupling $\lambda_{\mathbf{qv}}$ can be computed using the following expression,

$$\lambda_{\mathbf{qv}} = \frac{2}{\hbar N(0) N_{\mathbf{k}}} \sum_{nm\mathbf{k}} \frac{1}{\omega_{\mathbf{qv}}} |g_{\mathbf{k},\mathbf{qv}}^{nm}|^2 \delta(\varepsilon_{\mathbf{k}}^n) \delta(\varepsilon_{\mathbf{k}+\mathbf{q}}^m), \quad (1)$$

where $N(0)$ is the Fermi level density of states (DOS) of electrons, and $N_{\mathbf{k}}$ is the sum of $\mathbf{k}$ points in the fine k mesh. The phonon frequency is expressed as $\omega_{\mathbf{qv}}$. The EPC matrix element is denoted as $g_{\mathbf{k},\mathbf{qv}}^{nm}$. The indices of energy bands and phonon modes are written as (n, m) and ν . $\varepsilon_{\mathbf{k}}^n$ and $\varepsilon_{\mathbf{k}+\mathbf{q}}^m$ denote the Kohn-Sham orbital eigenvalues measured from the Fermi level.

The EPC constant λ can be calculated by summing $\lambda_{\mathbf{qv}}$ over the entire first Brillouin space or by integrating the Eliashberg spectral function $\alpha^2 F(\omega)$,

$$\lambda = \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{qv}} \lambda_{\mathbf{qv}} = 2 \int \frac{\alpha^2 F(\omega)}{\omega} d\omega, \quad (2)$$

where $N_{\mathbf{q}}$ denotes the total number of $\mathbf{q}$ irreducible points within the fine $\mathbf{q}$ mesh, and the Eliashberg spectral function $\alpha^2 F(\omega)$ is computed with

$$\alpha^2 F(\omega) = \frac{1}{2N_{\mathbf{q}}} \sum_{\mathbf{qv}} \lambda_{\mathbf{qv}} \omega_{\mathbf{qv}} \delta(\omega - \omega_{\mathbf{qv}}). \quad (3)$$

III. THE CRYSTAL STRUCTURE AND THERMODYNAMIC STABILITY

The InvDesFlow model generated the initial hydride crystal structures, upon which we conducted density functional theory (DFT) structural relaxation. As shown in Fig. 1, all multinary hydrides A_2BH_6 ($A = K$ and Na ; $B = Cu$ and Ag) and A_2XBH_6 ($A = K$ and Na ; $B = Cu$ and Ag ; $X = Ga$ and Li) in our research share the same space group of $Fm\bar{3}m$. A , B , and H atoms occupy $8c$ (0.25, 0.25, 0.25), $4a$ (0, 0, 0), and $24e$ (0.25, 0, 0) Wyckoff positions, respectively, where B locates at the center of an octahedron that is constructed by the six nearest hydrogen atoms. In quaternary hydride, the intercalated X atoms occupy the position of $4b$ (0.5, 0, 0). The detailed structures of all multinary hydrides are provided in SM Sec. III [34], and the relaxed structure files are available [47].

Our Invdesflow model integrates a discriminative component for predicting crystal thermodynamic stability. We benchmarked the reliability of Invdesflow model against DFT calculations, specifically for the forma-

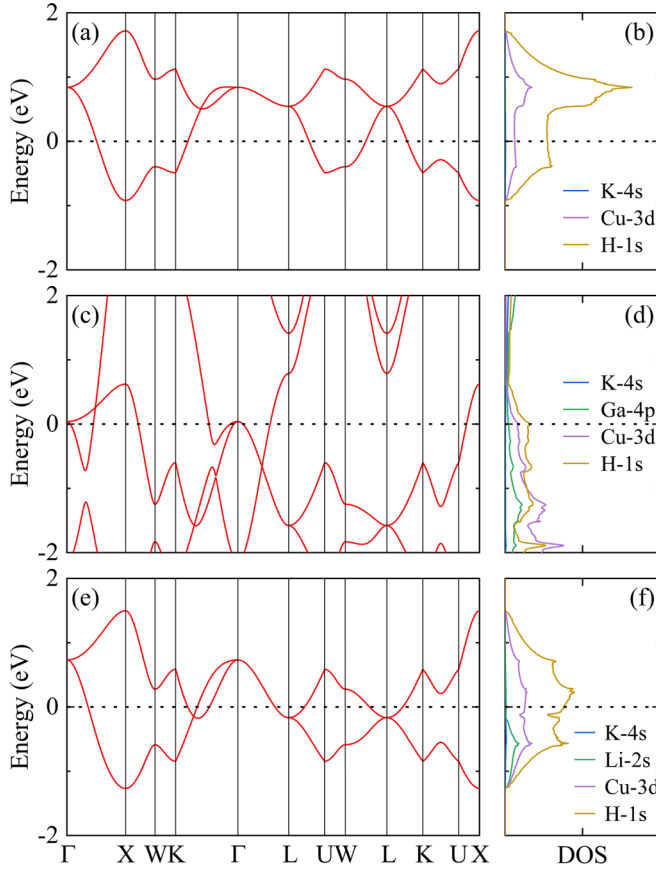


FIG. 2. Band structures and atomic orbital-projected density of states of (a), (b) K_2CuH_6 , (c), (d) K_2GaCuH_6 , and (e), (f) K_2LiCuH_6 , respectively. The Fermi level is set to zero.

tion energy of K_2GaCuH_6 . The investigated reaction, $6\text{KH} + \text{Cu} + \text{Ga} \rightarrow \text{K}_2\text{GaCuH}_6 + 4\text{K}$, shows that the formation energy of the left side is 0.137 eV/atom lower than that of the right side. Our InvDesFlow model predicted a formation energy difference of 0.146 eV/atom between the left and right sides of the synthetic pathway, which is in close agreement with the DFT-based reaction energy and forms the basis for all subsequent findings in this paper. The formation energies and the potential synthesis pathways for all multinary hydrides studied, as predicted by the InvDesFlow model, are available in SM Sec. III [34].

IV. RESULTS

Figure 2 shows the electronic structures of K_2CuH_6 , K_2GaCuH_6 , and K_2LiCuH_6 under ambient pressure. We see that K_2CuH_6 is a single-band metal, and K_2GaCuH_6 and K_2LiCuH_6 each have two bands crossing the Fermi level. Figures 2(d)–2(f) show the corresponding calculated projected density of states (PDOS). The bands near the Fermi level of these hydrides are primarily contributed by Cu-H octahedra, with negligible contributions from the K atom. A similar situation is also found in the intercalated atoms in the two quaternary hydrides, where Ga and Li atoms contribute little to the low-energy PDOS. Given that Cu is a transition metal with d orbitals, we therefore employed DFT + U cal-

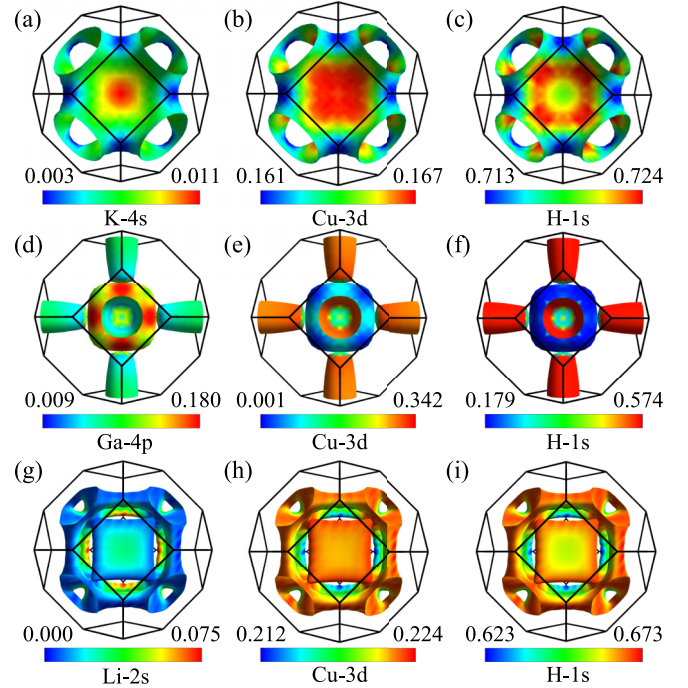


FIG. 3. The Fermi surfaces with colored representations of orbital-projected weights of (a)–(c) K_2CuH_6 , (d)–(f) K_2GaCuH_6 , and (g)–(i) K_2LiCuH_6 , respectively. The Contribution of the K atom is not shown because there the orbital-projected weight on the Fermi surface is weaker than that of the intercalated atom.

culations to investigate the influence of these orbitals on the electronic structure near the Fermi level of the K-Cu-H system (see SM Sec. IV [34]).

Figures 3(a)–3(c) show the orbital-projected Fermi surfaces of K_2CuH_6 . Since the electronegativity of the K atom is weaker than that of H and Cu, the K-4s orbital contributes little, and the electronic states on the Fermi surface mainly come from the Cu-H octahedron. Meanwhile, the H-1s orbital in the octahedron is dominant because to H atom is more electronegative. In quaternary hydrides [Figs. 3(d)–3(f) and Figs. 3(g)–3(i)]. Since Ga and Li are less electronegative than Cu and H, Ga and Li cannot compete with the octahedron in their ability to capture electrons; the Fermi surfaces of the two quaternary hydrides are still mainly contributed by the H-1s orbital and Cu-3d orbital. It is also worth noting that the distributions of the orbital-projected weights on the Fermi surfaces show significant anisotropy. For example, H atoms contribute largely to the trumpet-shaped Fermi surface but less to the cubic Fermi surface. According to Figs. 2 and 3, it can be seen that both the number of energy bands crossing the Fermi level and the shape of the Fermi surface change significantly, indicating that the intercalated atoms have altered the electronic structures. The detailed electronic structures of the Na-Cu-H and Na-Ag-H systems are provided in SM Sec. IV [34].

Next, we illustrate the phonon and EPC of the K-Cu-H system under ambient pressure. Figures 4(a)–4(c) show the phonon spectra of K_2CuH_6 , K_2GaCuH_6 , and K_2LiCuH_6 , respectively. There is no imaginary phonon mode under ambient pressure, which indicates that these multinary hydrides are

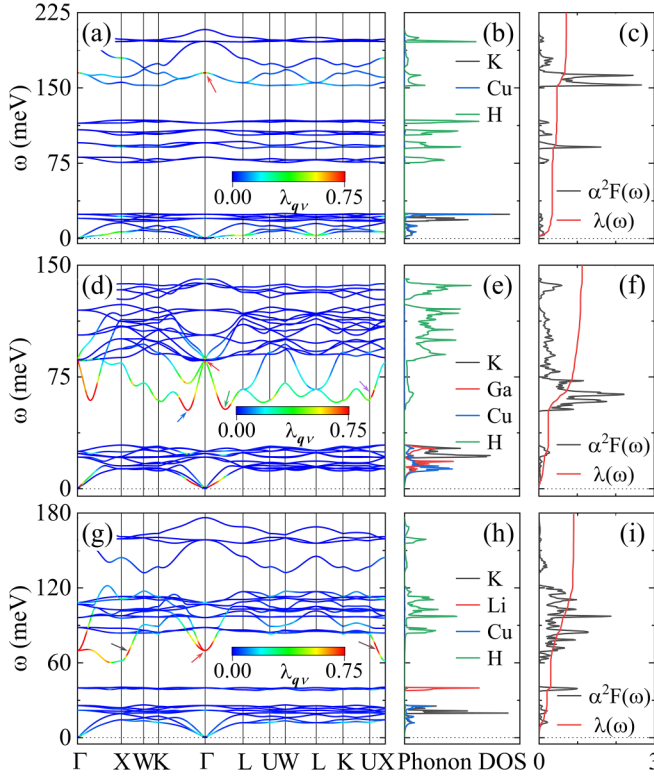


FIG. 4. Phonon spectra with colored representations of EPC strength λ_{qv} , weights, projected phonon DOS, Eliashberg spectral function $\alpha^2F(\omega)$, and accumulated $\lambda(\omega)$ of (a)–(c) K_2CuH_6 , (d)–(f) K_2GaCuH_6 , and (g)–(i) K_2LiCuH_6 under ambient pressure, respectively.

dynamically stable. Figure 4(d) shows the projected phonon DOS of K_2CuH_6 . The phonon spectrum of K_2CuH_6 can be divided into three regions. In the region of approximately less than 30 meV, the acoustic phonons and low-frequency optical branches are mainly contributed by the vibrations of K and Cu, and the rest of the high-energy phonons are totally contributed by the vibrations of H. As shown in Fig. 4(e), the phonon spectrum of K_2GaCuH_6 mainly has two components. In the region below 30 meV, it is mainly contributed to by mixed vibrations of K, Ga, and Cu, and the region above 50 meV is entirely contributed to by H. The projected phonon DOS of K_2LiCuH_6 is shown in Fig. 4(f), which has three major phonon modes contributing to it. In the region below 30 meV, phonons are mainly derived from the vibrations of K and Cu. Different from K_2GaCuH_6 , around 40 meV, the Li atom contributes to the phonon DOS alone because the atomic mass of Li is significantly lighter than that of K and Cu. In the region above 60 meV, it is entirely contributed by H.

Moreover, the colored representations in Figs. 4(a)–4(c) also show the distributions of the strength of the EPC of these hydrides. In the ternary hydride K_2CuH_6 , the strongest EPC is found in the E_g mode (165 meV) at the Γ point [indicated by the red arrow in Fig. 4(a)]. As shown in Fig. 4(b), after intercalating a Ga atom, the E_g mode at the Γ point still has relatively strong EPC [indicated by the red arrow in Fig. 4(b)], but the phonon energy decreases to about 85 meV. The intercalated Ga atom also induces more phonon modes with strong EPC,

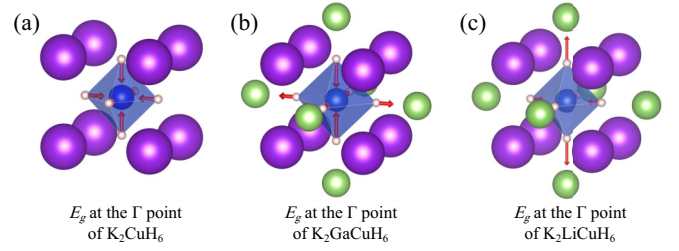


FIG. 5. The vibration modes of E_g at the Γ point of (a) K_2CuH_6 , (b) K_2GaCuH_6 , and (c) K_2LiCuH_6 , respectively. The purple, blue, and pink atoms denote K, Cu, and H atoms, respectively; the green atom denotes the intercalated Ga or Li atom.

such as the mode ~ 53 meV on the path from the Γ point to the K point that is marked by a blue arrow. Figure 4(c) shows that the intercalated Li atom significantly enhances the EPC strength of the E_g mode at the Γ point, and the frequency drops to about 70 meV [indicated by the red arrow in Fig. 4(c)]. Meanwhile, two new phonon modes with strong EPC appear. In addition, it can be seen that both the intercalations of Ga and Li cause phonon softening. Figures 5(a)–5(c) show the patterns of the E_g mode at the Γ point in the K-Cu-H system, and these modes with strong EPC are the breathing modes of the Cu-H octahedrons. Lattice distortion caused by intercalated atoms directly induces phonon softening in modes with strong EPC. In the K-Cu-H system, intercalated Ga and Li expand the Cu-H bond from 1.56 to 1.81 and 1.66 Å (SM Sec. III [34]), respectively, which varies the initial force equilibrium of the Cu-H octahedra. Hence, the modulations of vibrational modes are a natural outcome of this modified local environment. This is beneficial for the EPC constant λ [Eq. (2)], as low-frequency phonons enhance λ . Additionally, the intercalated atoms introduce extra vibrational degrees of freedom, diversifying the phonon spectrum and expanding the EPC phase space. This mechanism is anticipated to increase the number of strong-coupling modes.

As shown in Figs. 4(g)–4(i), the EPC constants λ of K_2CuH_6 , K_2GaCuH_6 , and K_2LiCuH_6 are calculated to be 0.71, 1.13, and 0.90, respectively. Figure 4(c) shows that the phonons below 160 meV have contributed more than 80% of the total EPC λ in K_2CuH_6 . By intercalating a Ga atom, the number of vibration modes with strong EPC increases, and these strong EPC modes ranging from 50 to 90 meV contribute approximately 64% of the total λ [Fig. 4(h)]. A similar situation is also found in the Li-intercalated case [Fig. 4(i)]. These results indicate that the intercalating atoms can induce more phonon modes with strong EPC or significantly enhance the EPC strength. In addition, we show the phonon and EPC calculation results of Na-Cu-H and Na-Ag-H systems in SM Sec. IV [34].

By solving anisotropic Eliashberg equations, we obtain the superconducting T_c of these hydrides. The superconducting gap Δ of the K-Cu-H system is shown in Fig. 6. The superconducting gap Δ of K_2CuH_6 disappears at 18 K, indicating that the superconducting T_c of K_2CuH_6 is 18 K under ambient pressure. As for K_2GaCuH_6 and K_2LiCuH_6 , the superconducting T_c is respectively calculated to be 68 and 53 K under ambient pressure [Figs. 6(b) and 6(c)], which is

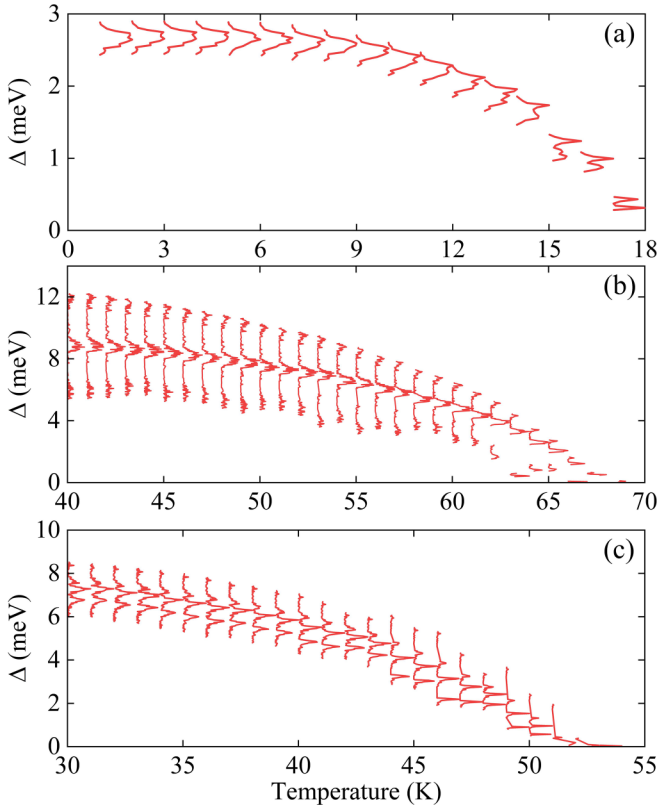


FIG. 6. Normalized anisotropic superconducting gap Δ of (a) K_2CuH_6 , (b) K_2GaCuH_6 , and (c) K_2LiCuH_6 under ambient pressure, respectively. The screening Coulomb potential μ^* is set to be 0.1.

significantly enhanced in comparison with that of the non-intercalated K_2CuH_6 . In addition, we further evaluated the temperature dependence of the isotropic superconducting gap Δ . The results demonstrate that the isotropic approximation significantly underestimates superconducting T_c . For instance, the calculated T_c for K_2GaCuH_6 and K_2LiCuH_6 decreases from 68 and 53 K (anisotropic) to 59 and 44 K (isotropic), respectively. For detailed data, see SM Sec. IV [34].

Figure 7 shows the calculated results of anisotropic superconductivity of the K-Cu-H system. In K_2GaCuH_6 , the EPC $\lambda(\mathbf{k})$ on the trumpet-shaped Fermi surface is relatively weak, while the EPC $\lambda(\mathbf{k})$ on the cubic edge is relatively strong [Fig. 3(d)]. The electron localization function (ELF) of the crystal plane shown here passes through the plane where the Cu and H atoms are located. Electron localization functions suggest that the valence electrons are mainly localized around the H atoms. These findings suggest that the vibrations of H atoms could significantly change electronic density and induce strong EPC. The relevant calculation results of anisotropic superconductivity of Na-Cu-H and Na-Ag-H systems are shown in the SM Sec. IV [34].

For the series of hydrides studied in this work, Table I summarizes the calculated EPC constant λ , logarithmic average phonon frequency $\omega_{\log}$, and superconducting transition temperature T_c . Note that $\text{Na}_2\text{GaAgH}_6$ is dynamically unstable and is included in the table for completeness. In the K-Cu-H system, the intercalated atoms successfully enhance the strength of EPC in the qua-

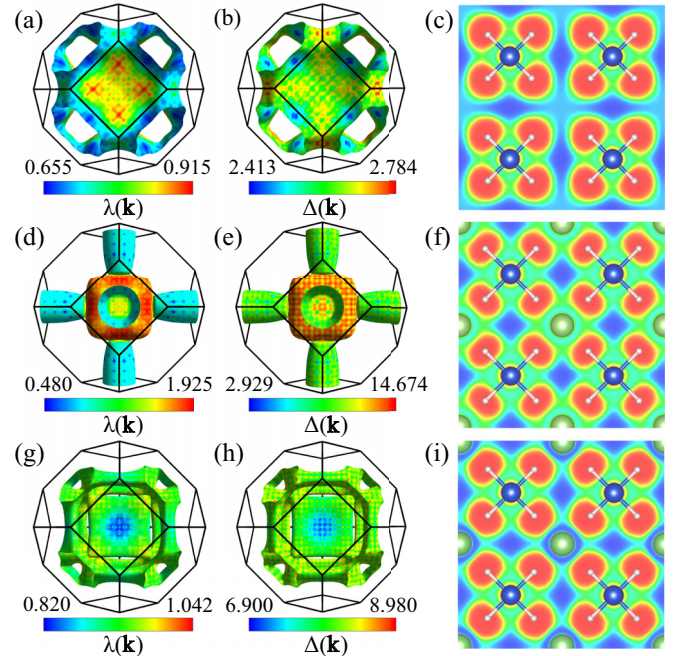


FIG. 7. Distributions of EPC $\lambda(\mathbf{k})$ and superconducting gap $\Delta(\mathbf{k})$ on the Fermi surfaces, and electron localization function of (a)–(c) K_2CuH_6 , (d)–(f) K_2GaCuH_6 , and (g)–(i) K_2LiCuH_6 , respectively. The distributions of the superconducting gap $\Delta(\mathbf{k})$ are calculated at 5 K. The blue and pink atoms denote Cu and H atoms, respectively; the green atom denotes the intercalated Ga or Li atom.

ternary hydride and significantly increase T_c . This indicates that the superconductivity of the crystal can be modulated by intercalating atoms, providing a feasible direction for the discovery of more potential high-temperature superconductors.

Our research further emphasizes that searching for materials with strong coupling modes is a promising path toward high- T_c superconductors. The Na-Ag-H system (Table I) exemplifies this, as its generally larger λ value results in a T_c significantly higher than those of the other hydride systems. Nevertheless, an intriguing exception occurs within this system: Li intercalation increases the EPC constant λ from 1.3213 to 1.6131, but the corresponding T_c decreases. This apparent anomaly can be understood through the

TABLE I. The EPC constant λ , logarithmic average phonon frequency $\omega_{\log}$, and superconducting transition temperature T_c by solving self-consistently the anisotropic Eliashberg equations of all multinary hydrides.

Formula	λ	$\omega_{\log}$	T_c (K)
K_2CuH_6	0.7120	29.5284	18
K_2GaCuH_6	1.1261	48.3532	68
K_2LiCuH_6	0.8995	54.7600	53
Na_2CuH_6	0.8699	61.7837	56
$\text{Na}_2\text{GaCuH}_6$	0.8164	62.9267	42
$\text{Na}_2\text{LiCuH}_6$	0.7305	65.1252	43
Na_2AgH_6	1.3213	50.4876	89
$\text{Na}_2\text{LiAgH}_6$	1.6131	42.2060	86
$\text{Na}_2\text{GaAgH}_6$			

McMillan-Allen-Dynes formula

$$T_c = \frac{\hbar\omega_{\log}}{1.2k_B} \exp \frac{-1.04(1 + \lambda)}{\lambda(1 - 0.62\mu^*) - \mu^*}, \quad (4)$$

where $\omega_{\log}$ is logarithmic average phonon frequency, and μ^* is the screening Coulomb potential, which dictates that T_c is not solely determined by λ but is also positively correlated with the logarithmic mean phonon frequency $\omega_{\log}$.

V. DISCUSSION

The strategy of intercalated atoms presents a viable method for inducing strong-coupling modes, offering a promising route for discovering high-temperature superconductors. However, the vast compositional space and high-dimensional phase space of multinary hydrides render their design exceptionally challenging. To address this complexity, we leveraged AI technology. Building upon the prior discovery of Li_2AuH_6 , we further explored the superconductivity of quaternary hydrides with the assistance of AI.

VI. CONCLUSION

In summary, combining machine learning and the first-principles theoretical calculations, we propose the atom-intercalated quaternary hydrides as high- T_c superconductor

candidates under ambient pressure. Specifically, the superconducting T_c of intercalated K_2GaCuH_6 and K_2LiCuH_6 are respectively predicted to be ~ 68 and 53 K under ambient pressure, which represents a significant increase compared to the 18 K of K_2CuH_6 . The study of EPC shows that intercalating an atom could lead to phonon softening and induce more phonon modes with strong EPC, which may be helpful for realizing high- T_c superconductivity. Our work provides a theoretical reference in the search for superconducting quaternary hydrides.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [35,47], embargo periods may apply.

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